

Explainable Customer Churn Prediction using Apache Spark & XAI

Complete Project Guide, Architecture & Execution Walkthrough

1. Project At a Glance & What It Does

This project builds an enterprise-grade Big Data and Explainable AI (XAI) platform for customer churn prediction in the telecommunications sector.

Traditional ML models are black-boxes that predict churn without explaining why. This system bridges that gap by combining Apache Spark for scalable data processing, Spark MLlib & ensemble learning (Gradient Boosted Trees with ROC-AUC = 0.8465), and game-theoretic SHAP interpretability to give both global macro-drivers and single-customer waterfall risk breakdowns.

Key Capabilities:

- Big Data Ingestion: Scalable schema inference & partitioned Parquet lake.
- Machine Learning: Multi-model benchmark (LR, RF, GBT) with 3-fold cross validation.
- Explainable AI: SHAP Beeswarm, Feature Importance & Local waterfall plots.
- Interactive Dashboard: Real-time single-customer risk scoring, batch CSV prediction engine, and executive PDF reporting.

2. Model Benchmarking Leaderboard

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Gradient Boosted Trees (Champion)	80.41%	0.6678	0.5214	0.5856	0.8465
Logistic Regression	80.62%	0.6678	0.5374	0.5956	0.8457
Random Forest Classifier	80.06%	0.6716	0.4866	0.5643	0.8436

3. How to Access & Run Locally

GitHub Repository: <https://github.com/Navneet1605/customer-churn-xai>

Step 1: Clone the repository

```
git clone https://github.com/Navneet1605/customer-churn-xai.git
cd customer-churn-xai
```

Step 2: Install dependencies

```
pip install -r requirements.txt
```

Step 3: Run the end-to-end pipeline (Ingestion -> Training -> SHAP -> Reports)

```
python run_pipeline.py
```

Step 4: Launch the interactive Streamlit Web Dashboard

```
python -m streamlit run dashboard/app.py
```

Access the web app at: <http://localhost:8501>